

Rethinking On-Policy Distillation of Large Language Models: Phenomenology, Mechanism, and Recipe

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Code: <https://github.com/thunlp/OPD>.

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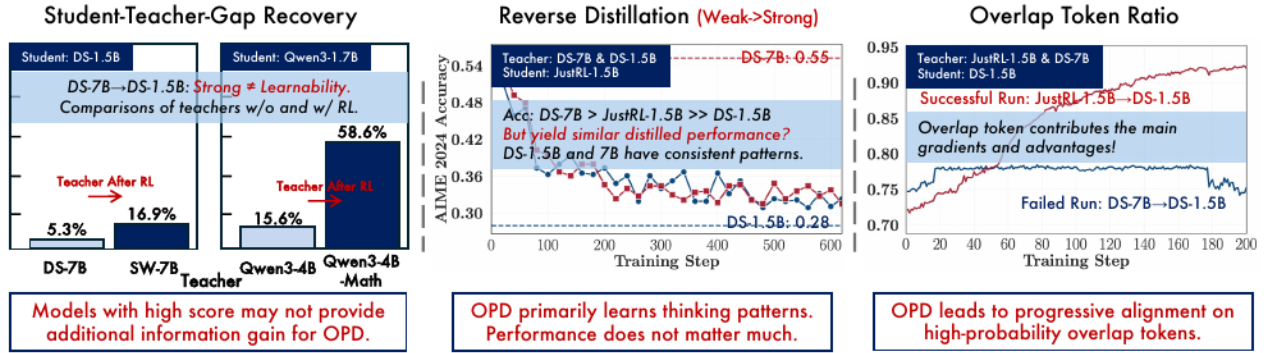


Figure 1 | Overview of our paper. JustRL-1.5B is obtained by applying RL to DeepSeek-Distill-1.5B (DS-1.5B), and Skywork-OR1-Math-7B (SW-7B) by applying RL to DeepSeek-Distill-7B (DS-7B).

Abstract | On-policy distillation (OPD) has become a core technique in the post-training of large language models, yet its training dynamics remain poorly understood. This paper provides a systematic investigation of OPD dynamics and mechanisms. We first identify that two conditions govern whether OPD succeeds or fails: (i) the student and teacher should share compatible thinking patterns; and (ii) even with consistent thinking patterns and higher scores, the teacher must offer genuinely new capabilities beyond what the student has seen during training. We validate these findings through weak-to-strong reverse distillation, showing that same-family 1.5B and 7B teachers are distributionally indistinguishable from the student’s perspective. Probing into the token-level mechanism, we show that successful OPD is characterized by progressive alignment on high-probability tokens at student-visited states, a small shared token set that concentrates most of the probability mass (97%–99%). We further propose two practical strategies to recover failing OPD: off-policy cold start and teacher-aligned prompt selection. Finally, we show that OPD’s apparent free lunch of dense token-level reward comes at a cost, raising the question of whether OPD can scale to long-horizon distillation.

1. Introduction

On-policy distillation (OPD) has rapidly emerged as a core technique for large language model (LLM) post-training. Recent industry efforts, including Qwen3 [Yang et al., 2025], MiMo [Xiao et al., 2026] and GLM-5 [Zeng et al., 2026], all adopt OPD in their post-training pipelines and report substantial gains, establishing it as a competitive complement to conventional supervised fine-tuning (SFT) and outcome-reward reinforcement learning (RL). Thinking Machines Lab [Lu and Lab, 2025] replicates the Qwen3 OPD recipe at a fraction of the RL compute cost, independently confirming that on-policy, dense supervision is a practically efficient alternative.